Sungjun Heo

Ulsan, Republic of Korea | Personal Website | Lab Website | Sungjun Heo

Education

Integrated M.S.–Ph.D. — **Ulsan National Institute of Science and Technology (UNIST)** — Mar 2024 -
- Graduate School of Artificial Intelligence
- Robotics & Mobility Laboratory of Prof.J Jeon
- Research Field : End-to-End Autonomous Driving, Reinforcement Learning based Planning, Robotics.
- GPA : 4.0 / 4.3

B.S — **Ulsan National Institute of Science and Technology (UNIST)** — Mar 2021 - Fed 2024
- Department of Electrical Engineering
- Early Graduation (by one year)
- GPA : 3.6 / 4.3 , Total credits : 127

HS — **Changwon Science High School (CSHS)** — 2018 - 2020

Research Interest

Autonomous Driving (AD)
- End-to-End (E2E) driving, Trajectory planning, Spatiotemporal fusion model

Robot manipulation
- Learning based manipulation, Multi-Agent, Data aggregation

Reinforcement Learning (RL)
- Hierarchical RL, Curriculum RL, Multi-Agent RL

Publications

2025 — **Sungjun Heo** and J. Jeon, **"Predicted State-Based Hierarchical Reinforcement Learning for Long-Term Decision Making in Urban Dynamic Scenarios"** - Proceeding in IEEE Intelligent Vehicles Symposium (Flagship Conference of IEEE ITSS - IV 2025), pp. 1164–1170.

2026 — H. Jeong, E. Kim, **Sungjun Heo**, S. Kim, S. Lee, J. Shin et al., **"Deployment-Oriented End-to-End Autonomous Driving: Enhancing Closed-Loop Stability with a Lightweight Framework"** - Proceeding in IEEE Intelligent Vehicles Symposium (Flagship Conference of IEEE ITSS - IV 2026), pp. 694-701. / **(Oral Presentation, Top 8.5% paper)**

2026 — **Sungjun Heo** and J. Jeon, **"SHIFT-RL: Sensor-driven Hierarchical Information Fusion Transformer for BEV-based Maneuvering in Dense Multi-lane Environments "** - Accepted; to be presented at IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS), (Major Conference of IEEE RAS - IROS 2026)

2026 — **Sungjun Heo**, H. Jeong, E. Kim, H. Lee, J. Lee, S. Lee, et al., **"Capacity-Controlled Enforcement of Sparse Traffic-Light Cues for Rule-Conformant End-to-End Driving"** - Accepted; to be presented at IEEE International Conference on Systems, Man, and Cybernetics (SMC), (Flagship conference of the IEEE SMC Society - 2026 SMC) / **(Oral Presentation)**

2026 — **Sungjun Heo** and J. Jeon, **"Decoupling Actor and Critic Replay for Offline Reinforcement Learning with Three-Factor Selection"** - Accepted; to be presented at IEEE International Conference on Systems, Man, and Cybernetics (SMC), (Flagship conference of the IEEE SMC Society - 2026 SMC) / **(Oral Presentation)**

2026 — **Sungjun Heo**, S. Yang, H. Jeong, J. Ha, J. Jeon **"To be disclosed after the double-blind peer review process"**

Awards and Achievements

Academic Achievement Award 2021

- Undergraduate academic achievement excellence award in 2021

Undergraduate Research Award (Top 5 / All graduation students) Dec 2023

- Research Title : Autonomous Driving Waypoint-generation based on Vehicle Kinematics.

3rd place in Preliminary Round (3 / 16 teams) Mar 2025
- Awarded $3,600

- Preliminary Round of 2025 Hyundai Motor Group's Autonomous Driving Challenge
- RGB camera based End-to-End Mission-based Autonomous driving challenge in K-City

1st place in 2025 Hyundai Motor Group's Autonomous Driving Challenge (1 / 16 teams) Sep 2025
- Awarded $22,000 + (Special privilege) Autonomous driving field trip

- Final 2025 Hyundai Motor Group's Autonomous Driving Challenge
- RGB camera based End-to-End Full-Self-Driving in K-City

Excellence Award | AI Tech Open Workshop Oct 2025
- Awarded $2,400

- UNIST AI Tech Open Workshop | Team FAST
- Research title : Development of an End-to-End Autonomous Driving Model Based on a High-Dimensional Simulator

Grand Prize - Brain To Society, U-Challenge Festival (1 / 96 teams) Dec 2025
- Awarded $3,600

- Brain To Society : Future Mobility part
- Title : Sensor-driven Hierarchical Information Fusion Transformer for BEV-based Maneuvering in Dense Multi-lane Environments

IEEE SMC 2026 Student and Young Professional Travel Grant Oct 2026
- Awarded $600

- IEEE Systems, Man, and Cybernetics Society

Work Experience

Paper Reviewer

- 3 Papers in 2025 IEEE INTELLIGENT VEHICLES SYMPOSIUM
- 2 Papers in 2025 IEEE International Conference on Intelligent Transportation Systems
- 2 Papers in 2026 IEEE INTELLIGENT VEHICLES SYMPOSIUM
- 2 Paper in 2026 IEEE/RSJ International Conference on INTELLIGENT ROBOTS & SYSTEMS

AI Innovation Park Internship Jul - Sep 2024

- Development of a time-series-based AI model for power prediction

Project

• Major Projects & Challenge

Preliminary Round : 2025 Hyundai Motor Group's Autonomous Driving Challenge Sep 2024 - Mar 2025

- Position : Team Member(model framework, dataset)
- RGB camera based End-to-End Mission-based Autonomous driving challenge in K-City

Final Round : 2025 Hyundai Motor Group's Autonomous Driving Challenge Apr 2025 - Oct 2025

- Position : Team Member(rule-based expert driving dataset, model framework)
- RGB camera based End-to-End Full-Self-Driving in K-City

2025 Brain to Society : Future mobility Jul - Dec 2025

- Position : Team Leader
- Project Title : Sensor-driven Hierarchical Information Fusion Transformer for BEV-based Maneuvering in Dense Multi-lane Environments

2026 Brain to Society : Future mobility Jun - Dec 2026

- Position : Team Leader
- Project Title : Capacity-Controlled Enforcement of Sparse Traffic-Light Cues for Rule-Conformant End-to-End Driving

• Academic Projects

Low-cost Robot Manipulation (Robotics: Science and Systems) Dec 2024

- Position : Team Member
- Paper Title :Demonstrating a Vision-Based AI Robot for Strategic Board Games
- Project page: link

3D Vision and Machine Cognition Jun 2025

- Position : Team Member
- Project Title : Traffic-Light and Boundary-Aware BEV Learning from Multi-Camera Images

2D Vision: Human Pose Estimation Jun 2025

- Title : SoftEdge-PoseContrast View Conditioned Human-Pose Classifier with Contrastive Learning